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C. Strategic Projects

Ashley Park - Wansley 500kV Line

TEAMS 21062

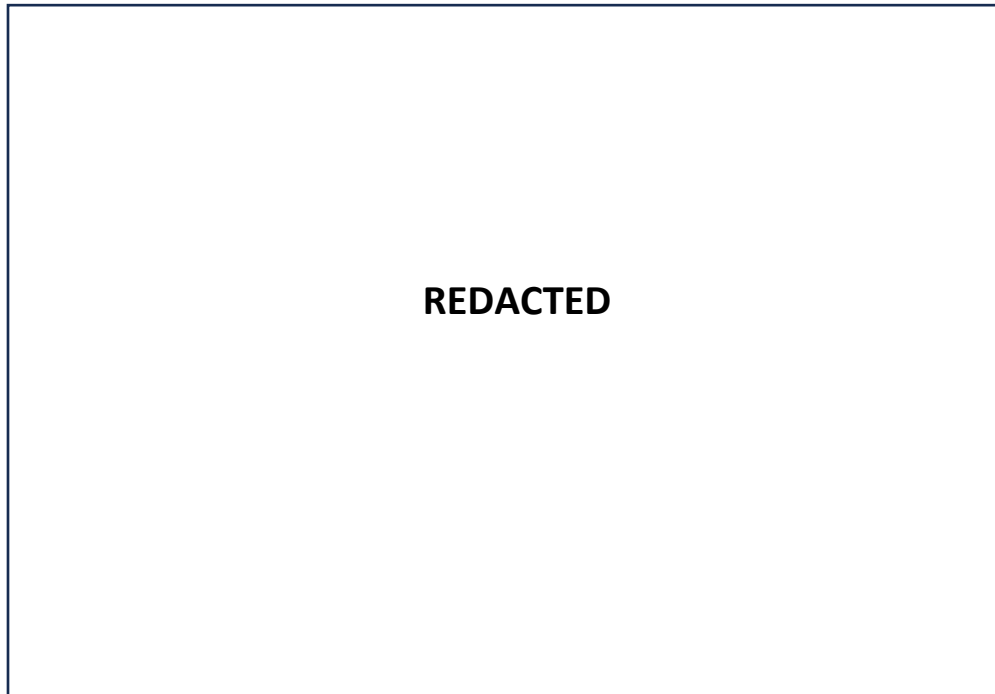
Project Need Date: June 1, 2029

Executive Summary

The construction of the Ashley Park – Wansley 500kV line aims to address the evolving dynamics within the Georgia Integrated Transmission System (ITS), primarily driven by the changes in generation and forecasted load growth. This project involves the construction of a new ~30-mile 500 kV line from Ashley Park to Wansley as shown in Figure 4. Additionally, supplemental projects include Ashley Park Line Termination (GPC) and Wansley Line Termination (MEAG). This project will be done in conjunction with GTC: Tenaska – Wansley 500kV (TEAMS 21123).

The decision to undertake this project stems from the necessity to enhance available transmission capacity and mitigate thermal limits resulting from 500 kV contingencies under NERC TPL-001-5. This project will reduce flows and increase available capacity on various circuits, thereby alleviating or reducing thermal constraints on critical circuits under contingency scenarios. This solution is the result of joint planning efforts amongst the GA ITS Participants.

Figure 4 Area System Map



Compliance Statement

This project addresses problems associated with Category P1 events. These problems were identified as part of Southern Company's Transmission Planning process in compliance with NERC Standard TPL-001-5.

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Background and Problem Description

The Ashley Park – Wansley 500kV line, in conjunction with the construction of the new GTC: Tenaska – Wansley 500kV line, effectively reduces flows and increases available capacity on overloaded circuits shown in Table 17. Table 18 indicates capacity increases on non-overloaded facilities. Both tables show capacity increases in relation to existing facility ratings.

Table 17 Load Flow Reductions on Overloaded Facilities

Facility Name	Facility Rating (MVA)	Loading Reduction
O'Hara - Charles 230kV	REDACTED	REDACTED
Loopers Farm - South Dalton	REDACTED	REDACTED
Villa Rica - Wansley 500kV	REDACTED	REDACTED
Dorsett - Thomaston 230kV	REDACTED	REDACTED
Bonaire Primary - Dorsett 230kV	REDACTED	REDACTED

Table 18 Capacity Increases on Non-Overloaded Facilities

Facility Name	Facility Rating (MVA)	Available Capacity Increase (MVA)	Loading Reduction
O'Hara - Ashley Park 500kV	REDACTED	REDACTED	REDACTED
O'Hara - Dresden 500kV	REDACTED	REDACTED	REDACTED
Dresden - Heard County 500kV	REDACTED	REDACTED	REDACTED
Heard County - Wansley 500kV	REDACTED	REDACTED	REDACTED
Heard County - Tenaska 500kV	REDACTED	REDACTED	REDACTED

In addition to solving thermal constraints and increasing capacity, the Ashley Park - Wansley 500kV along with the GTC: Tenaska – Wansley 500kV line provides improved voltage profiles for the west Georgia and greater Atlanta areas. Figures 5, 6, and 7 show the voltage difference on 500kV buses in the region with the addition of all new 500kV projects.

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Figure 5 Voltage Improvements-Big Shanty

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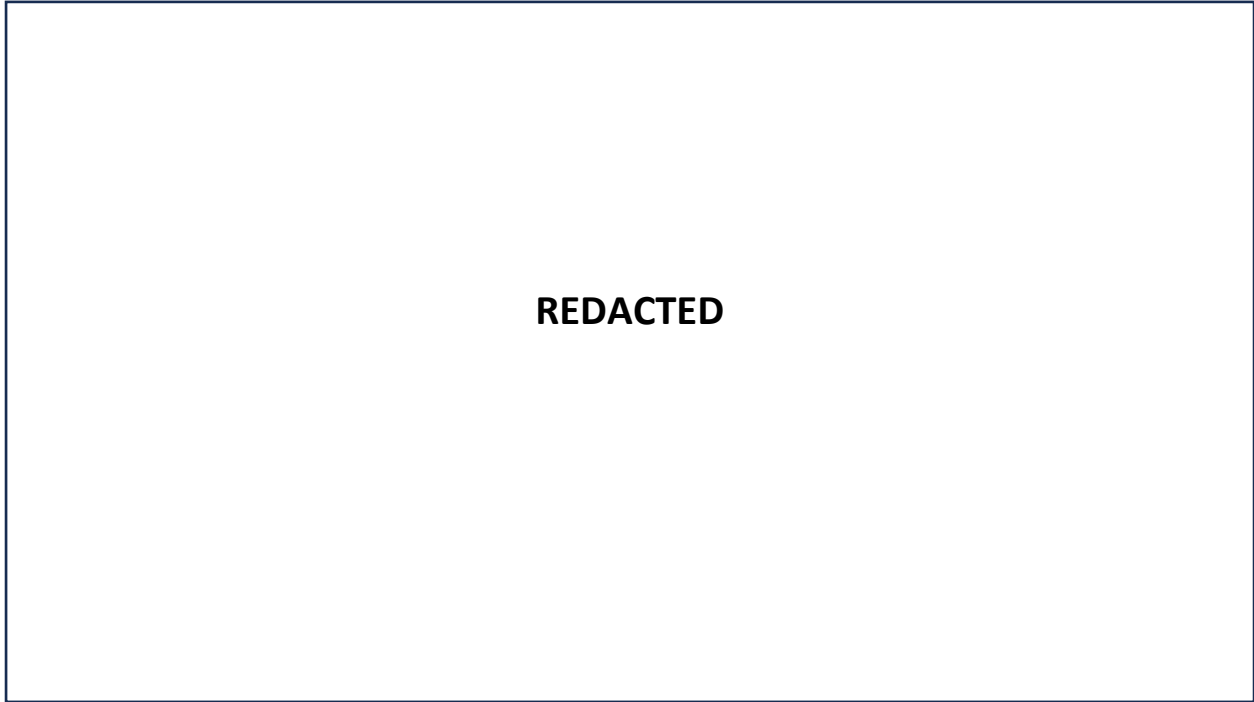
Figure 6 Voltage Improvements-Ohara

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Figure 7 Voltage Improvements-Tomochichi



Study Assumptions

- Load Flow Cases: 2024 series V2
- Study Years: 2028 – 2034
- Case Seasons: Summer Peak, Off-Peak, Daylight Shoulder, Dusk Shoulder

Discussion of Alternatives

The Preferred Plan (\$_{REDACTED}) consists of:

- Construction of the Ashley Park – Wansley 500 kV Line (~30 miles) with (3) 100C 1113 ACSR conductor.
- Expand Ashley Park and Wansley Primary 500 kV yards to connect & terminate the new line.
- Installation of communication equipment and integration with EMS.

The preferred plan accomplishes the following:

1. **High-Capacity Corridor:** Establishes a new 500 kV line from Ashley Park to Wansley, enhancing system capacity in the metro area.
2. **Reduced Flows, Increased Capacity:** Decreases loading and increases available capacity on multiple circuits, improving system reliability.
3. **Supports Generation Expansion:** Facilitates integration of additional generation at existing facilities in area, promotes sustainability, and grid resilience.
4. **Operational Flexibility:** Provides significantly improved operational flexibility and reliability for system outages and future maintenance activities on existing facilities, particularly on the critical 500 kV system.

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Alternate Plans

- **Alternate Plan 1: Do Nothing (\$REDACTED).** This is not a feasible option to remain in compliance with NERC Standard TPL-001-5. In addition to the TPL compliance implications of this option, the severity and breadth of identified system constraints would result in a severe degradation of service to Georgia Power customers.
- **Alternate Plan 2: Brute Force (\$REDACTED).** This involves the direct replacement or rebuild of the facilities listed in Table 19.

Table 19 Brute Force Estimate

Facility Name	Mileage	Planning Grade Estimate
O'Hara - Charles 230kV Rebuild	14	REDACTED
Loopers Farm – South Dalton Jumper	N/A	REDACTED
Villa Rica - Wansley 500kV Rebuild	26	REDACTED
Dorsett – Thomaston 230kV Rebuild	39	REDACTED
Bonaire Primary – Dorsett 230kV Rebuild	17.6	REDACTED
Total	97	REDACTED

Alternate Plan 2 is not a feasible option given the following factors:

1. Accommodating and coordinating necessary outages to facilitate maintenance of existing facilities and the construction activities for system improvements
2. Timing and reduced reliability exposure to system given length of corridors

Additionally, the Brute Force alternative estimates do not reflect the full scope of all necessary work and costs that would be required along with any additional generation redispatch impacts associated with the numerous required outages.

Conclusion and Recommendations

The Ashley Park – Wansley 500kV New Line was selected because it solves identified thermal constraints, provides additional 500kV corridor for generation power transfers, and minimizes outage impacts to the transmission system during construction.

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Farley (APC) – Tazewell 500kV Line

TEAMS 21063

Project Need Date: June 1, 2030

Executive Summary

The construction of the Farley (APC) – Tazewell 500 kV line aims to address the evolving dynamics within the Georgia Integrated Transmission System (ITS), primarily driven by changes in generation and forecasted load growth. This project involves the construction of a new ~120-mile 500 kV line from Farley (APC) to Tazewell as shown in Figure 8. Additionally, supplemental projects include Farley Line Termination (APC) and expansion of existing Tazewell 230kV station for new 500kV yard. This project will be done in conjunction with the GTC: Talbot #2 – Tazewell 500kV project (TEAMS 21076).

The decision to undertake this project stems from the necessity to enhance available transmission capacity and mitigate thermal limits resulting from 500 kV contingencies under NERC TPL-001-5. This project will reduce flows and increase available capacity on various circuits, thereby alleviating or reducing thermal constraints on critical circuits under contingency scenarios. This solution is the result of joint planning efforts amongst the GA ITS Participants.

Figure 8 Area System Map

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Compliance Statement

This project addresses problems associated with Category P1 events. These problems were identified as part of Southern Company's Transmission Planning process in compliance with NERC Standard TPL-001-5.

Background and Problem Description

The Farley - Tazewell 500 kV line, in conjunction with the construction of the new GTC: Talbot #2 – Tazewell 500kV line, effectively reduces flows and increases available capacity on overloaded circuits shown in Table 20. Table 21 indicates capacity increases on non-overloaded facilities. Both tables show capacity increases in relation to existing facility ratings.

Table 20 Low Flow Reductions on Overloaded Facilities

Facility Name	Facility Rating (MVA)	Loading Reduction
Raccoon Creek 500/230kV Autotransformer A	REDACTED	REDACTED
Cotton Primary - Scooter 230kV	REDACTED	REDACTED
Daisy - West Valdosta 230kV	REDACTED	REDACTED
Mitchell - Raccoon Creek 230kV	REDACTED	REDACTED
Raccoon Creek - Scooter 230kV	REDACTED	REDACTED
Kawikee - South Columbus 115kV	REDACTED	REDACTED
Moultrie - Thomasville 115V	REDACTED	REDACTED
Albany Primary - Palmyra 115kV	REDACTED	REDACTED
Blakely Primary - Lower River Road 115kV	REDACTED	REDACTED
Lizard Lope - Arlington Primary 115kV	REDACTED	REDACTED
Fortson - Talbot #2 230kV	REDACTED	REDACTED
Fortson - Goat Rock 230kV (Black)	REDACTED	REDACTED
Fortson - Goat Rock 230kV (White)	REDACTED	REDACTED
Americus - Thrill Hill 115kV	REDACTED	REDACTED

Table 21 Capacity Increases on Non-Overloaded Facilities

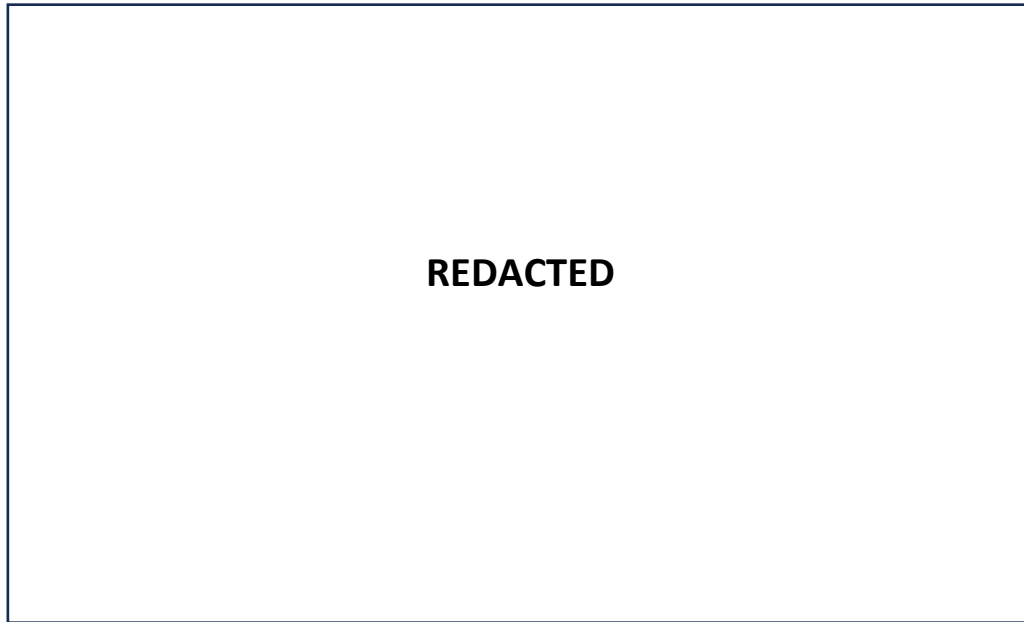
Facility Name	Facility Rating (MVA)	Available Capacity Increase (MVA)	Loading Reduction
Arlington Primary - Giles 115kV	REDACTED	REDACTED	REDACTED
Cuthbert Primary - Kawikee 115kV	REDACTED	REDACTED	REDACTED

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In addition to solving thermal constraints and increasing capacity, the Farley – Tazewell 500kV line along with the GTC: Talbot #2 – Tazewell 500kV line provides improved voltage profiles for the southwest and west Georgia areas. Figure 9 shows the improved voltage profile on key 500kV bus in southwest Georgia.

Figure 9 Voltage Profile



Study Assumptions

- Load Flow Cases: 2024 series V2
- Study Years: 2028 – 2034
- Case Seasons: Summer Peak, Off-Peak, Daylight Shoulder, Dusk Shoulder

Discussion of Alternatives

The Preferred Plan (\$REDACTED) consists of:

- Construction of the Farley (APC) – Tazewell 500 kV Line (~120 miles) with (3) 100C 1113 ACSR conductor.
- Expand existing Tazewell 230kV yard to install new 500/230kV autotransformer and create new 5 breaker 500kV ring. 500kV Ring will have line terminations to Farley (APC), Talbot #2, Blacksmith, and Talbot.
- Terminate 500kV line at Farley (APC)
- Installation of communication equipment and integration with EMS.

The preferred plan accomplishes the following:

1. **High-Capacity Corridor:** Establishes a new 500 kV line from Farley to Tazewell, enhancing system capacity from South to North.
2. **Reduced Flows, Increased Capacity:** Decreases loading and increases available capacity on multiple circuits, improving system reliability.
3. **Supports Generation Expansion:** Facilitates integration of additional generation at existing facilities along with new solar facilities, promotes sustainability, and grid resilience.
4. **Operational Flexibility:** Provides significantly improved operational flexibility and reliability for system outages and future maintenance activities on existing facilities, particularly on the critical 500 kV system.

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Alternate Plans

- **Alternate Plan 1:** Do nothing (\$_{REDACTED}). This is not a viable option to remain in compliance with NERC TPL-001-5. In addition to the TPL compliance implications of this option, the severity and breadth of identified system constraints would result in a severe degradation of service to Georgia Power customers.
- **Alternate Plan 2:** Brute Force (\$_{REDACTED}). This involves the direct replacement or rebuild of the facilities listed in Table 22.

Table 22 Brute Force Estimates

Facility Name	Rebuild Line Length (Miles)	Planning Grade Estimate
Raccoon Creek 500/230kV Autotransformer A	N/A	REDACTED
Cotton Primary - Scooter 230kV	7.25	REDACTED
Daisy - West Valdosta 230kV	15.9	REDACTED
Mitchell - Raccoon Creek 230kV	7.68	REDACTED
Raccoon Creek - Scooter 230kV	8.21	REDACTED
Kawikee - South Columbus 115kV	9	REDACTED
Moultrie - Thomasville 115V	13.1	REDACTED
Albany Primary - Palmyra 115kV	1.92	REDACTED
Blakely Primary - Lower River Road 115kV	9.5	REDACTED
Lizard Lope - Arlington Primary 115kV	21.6	REDACTED
Fortson - Talbot #2 230kV	14	REDACTED
Fortson - Goat Rock 230kV (Black)	9.9	REDACTED
Fortson - Goat Rock 230kV (White)	10	REDACTED
Americus - Thrill Hill 115kV	19.11	REDACTED
Total	147.17	REDACTED

Alternate Plan 2 is not a feasible option given the following factors:

1. Accommodating and coordinating necessary outages to facilitate maintenance of existing facilities and the construction activities for system improvements
2. Timing and reduced reliability exposure to system given length of corridors
3. Lack of solution robustness to facilitate power transfers from south Georgia to north Georgia driven by increased solar generation and load growth

Additionally, the Brute Force alternative estimates do not reflect the full scope of all necessary work and costs that would be required along with any additional generation redispatch impacts associated with the numerous required outages.

Conclusion and Recommendations

Although the Preferred Plan is not the least cost plan, it was selected because it solves identified thermal constraints, provides additional 500kV corridor for solar generation power transfers, and minimizes outage impacts to the transmission system during construction.

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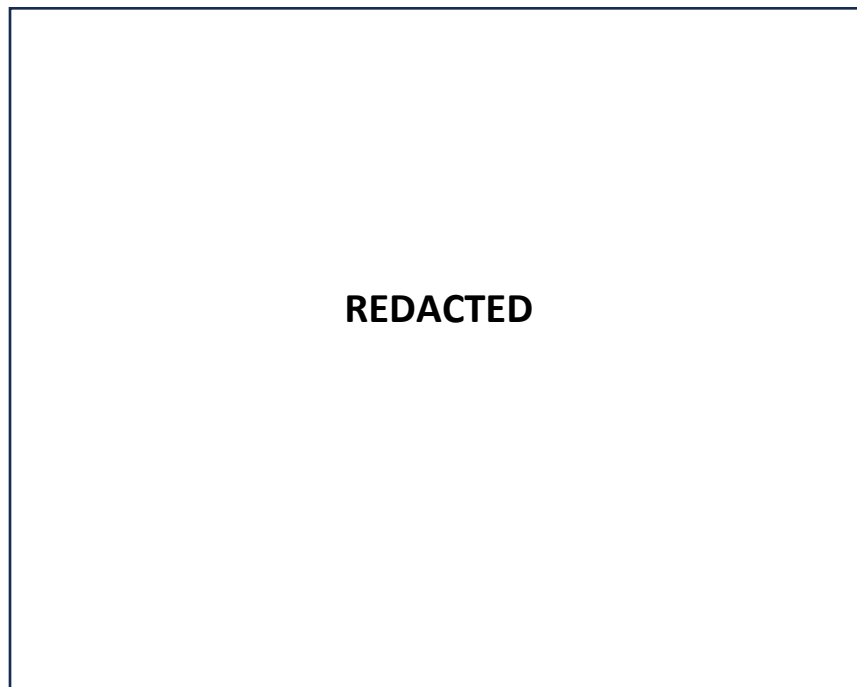
Hatch – Wadley 500kV Line TEAMS 20756 Project Need Date: June 1, 2031

Executive Summary

The construction of the Hatch – Wadley Primary 500 kV Line aims to address the evolving dynamics within the Georgia Integrated Transmission System (ITS), primarily driven by the increasing penetration of renewable generation plants and forecasted load growth. This project involves the construction of a new ~65-mile 500 kV line from Hatch to Wadley Primary as shown in Figure 10. Additionally, supplemental projects include Hatch Line Termination (GPC) and Wadley Primary Line Termination (MEAG).

The decision to undertake this project stems from the necessity to enhance available transmission capacity and mitigate potential thermal limits resulting from 500 kV contingencies under solar expansion scenarios. The 2023 ITS solar sensitivity study identified numerous benefits associated with the Hatch – Wadley 500 kV line, including providing an additional high-capacity corridor from South to North for solar penetration level of **REDACTED**. Furthermore, this project is anticipated to reduce flows and increase available capacity on various circuits, thereby alleviating or reducing thermal constraints on critical circuits under solar expansion scenarios. This solution is the result of joint planning efforts amongst the GA ITS Participants.

Figure 10 Area System Map



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Compliance Statement

This project addresses problems associated with Category P1 events. These problems were identified as part of Southern Company's Transmission Planning process in compliance with NERC Standard TPL-001-5.

Background and Problem Description

The Hatch – Wadley 500 kV line effectively reduces flows and increases available capacity on the below circuits. Table 23 indicates the reduction in percent MVA of existing flow on the lines. Table 24 indicates capacity increases in percent MVA and total MVA in relation to existing line ratings. Figure 11 provides a regional overview highlighting some of the impacted areas experiencing flow reductions and capacity increases.

Table 23 Load Flow Reductions

Monitored Facility	Rating (MVA)	Flow reductions (% MVA)
Bonaire Primary - Kathleen 230 kV	REDACTED	REDACTED
Branch – Glenwood Springs 230 kV	REDACTED	REDACTED
Branch - Tiger Creek (Black) 230 kV	REDACTED	REDACTED
Branch - Tiger Creek (White) 230 kV	REDACTED	REDACTED
Branch - Warrenton Primary 230 kV	REDACTED	REDACTED
Eatonton Primary - Porterdales Primary 230 kV	REDACTED	REDACTED
Hatch - Vidalia 230 kV	REDACTED	REDACTED
Hatch 500/230 kV Bank #10	REDACTED	REDACTED
Kathleen - Pitts 230 kV	REDACTED	REDACTED
North Dublin - Wadley Primary 230 kV	REDACTED	REDACTED
North Tifton - Pitts 230 kV	REDACTED	REDACTED
Sandersville #1 - Wadley Primary 230 kV	REDACTED	REDACTED
Thomson 500/230 kV Bank D	REDACTED	REDACTED
Tiger Creek – Wadley Primary 230 kV	REDACTED	REDACTED
Vidalia - Wadley Primary (White) 230 kV	REDACTED	REDACTED
Vidalia 230 kV Bus Tie	REDACTED	REDACTED
Wadley 500/230 kV Bank C	REDACTED	REDACTED
Wadley 500/230 kV LS Breaker #1	REDACTED	REDACTED
Wadley 500/230 kV LS Breaker #2	REDACTED	REDACTED
Mosley – Wadley 230 kV	REDACTED	REDACTED
Mosley – Vidalia 230 kV	REDACTED	REDACTED

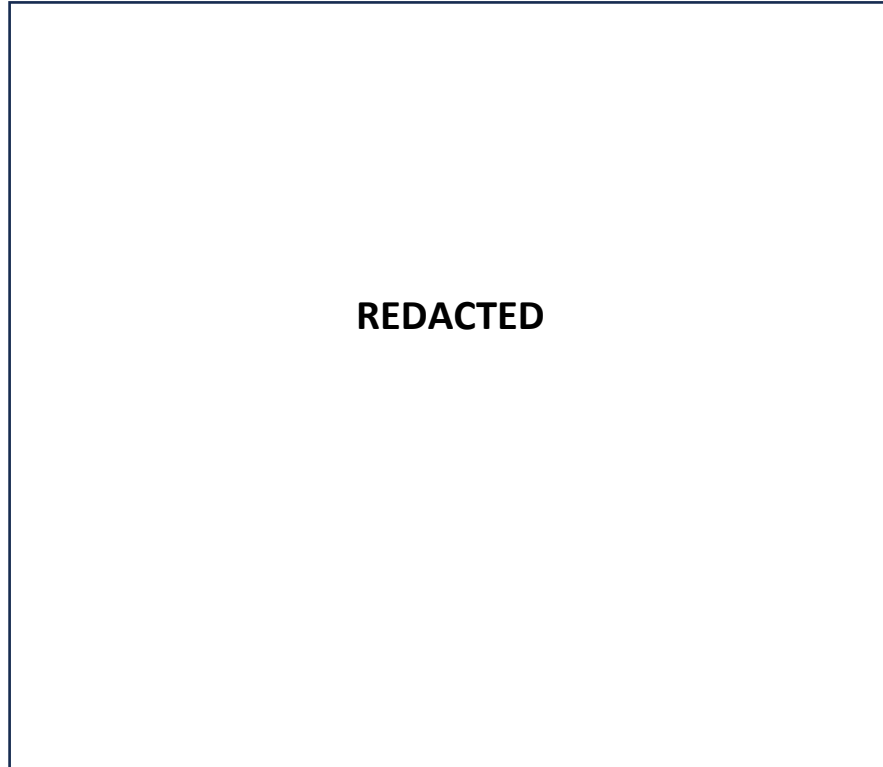
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Table 24 Capacity Increases

Monitored Facility	Rating (MVA)	Increase in Available Capacity (% MVA)	Increase in Available Capacity (MVA)
Hatch - Vidalia 230 kV	REDACTED	REDACTED	REDACTED
Hatch 500/230 kV Bank #10	REDACTED	REDACTED	REDACTED
Thomson 500/230 kV Bank D	REDACTED	REDACTED	REDACTED
Tiger Creek - Wadley Primary 230 kV	REDACTED	REDACTED	REDACTED
Vidalia - Wadley Primary (White) 230 kV	REDACTED	REDACTED	REDACTED
Vidalia 230 kV Bus Tie	REDACTED	REDACTED	REDACTED
Wadley 500/230 kV Bank C	REDACTED	REDACTED	REDACTED
Wadley 500/230 kV LS Breaker #1	REDACTED	REDACTED	REDACTED
Wadley 500/230 kV LS Breaker #2	REDACTED	REDACTED	REDACTED
Mosley – Wadley 230 kV	REDACTED	REDACTED	REDACTED
Mosley – Vidalia 230 kV	REDACTED	REDACTED	REDACTED

Figure 11 Impacted Area



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Study Assumptions

- Load Flow Cases: 2023 series V1 and V2
- Study Years: 2028 – 2033
- Case Seasons: Summer Peak, Off-Peak, Daylight Shoulder, Dusk Shoulder

Discussion of Alternatives

The Preferred Plan (\$REDACTED) consists of:

- Construction of the Hatch – Wadley Primary 500 kV Line (~65 miles) with (3) 100C 1113 ACSR conductor.
- Expand Hatch and Wadley Primary 500 kV yards to connect & terminate the new line.
- Installation of communication equipment and integration with EMS.

The preferred plan accomplishes the following:

1. **High-Capacity Corridor:** Establishes a new 500 kV line from Hatch to Wadley Primary, enhancing system capacity from South to North.
2. **Reduced Flows, Increased Capacity:** Decreases loading and increases available capacity on multiple circuits, improving system reliability.
3. **Supports Solar Expansion:** Facilitates integration of solar generation, promotes sustainability, and grid resilience.
4. **Operational Flexibility:** Provides significantly improved operational flexibility and reliability for system outages and future maintenance activities on existing facilities.

Alternate Plans

- **Alternate Plan 1:** Do nothing (\$REDACTED). The trend of increasing penetration of renewable (solar) generation plants on the Georgia Integrated Transmission System (ITS), and the resulting displacement of traditional fossil-fueled resources (particularly older coal facilities) is changing flow patterns on the system. These changes in addition to evolving interregional flow patterns and forecasted load growth are resulting in transmission congestion. If left unaddressed the system will not be able to facilitate future growth and maintain reliability requirements. This will introduce potential delays and limits until transmission improvements are identified and completed on an individual basis at the time of interconnection.

Conclusion and Recommendations

The Hatch – Wadley Primary 500 kV line project represents a strategic investment in the future reliability and efficiency of the Georgia Integrated Transmission System (ITS). Given the anticipated benefits, the GA ITS Solar Sensitivity Study team recommends proceeding with the implementation of the Hatch – Wadley 500kV project as part of the phased approach outlined in the 2023 Solar Sensitivity Analysis Results and Recommendations as presented to the Joint Committee on January 25, 2024.

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McGrau Ford – Middle Fork 500kV Line

TEAMS 09661

Project Need Date: June 1, 2033

Executive Summary

The construction of the McGrau Ford - Middle Fork 500kV transmission line is intended to address the unprecedented, forecasted load growth and the increase of solar penetration in the state of Georgia. The project builds a new 65 miles of 500kV line from Middle Fork to McGrau Ford as shown in Figure 12. Additionally, supplemental projects include the expansion of the 500kV yard at McGrau Ford (GPC) and the construction of a new 500kV yard at Middle Fork (GTC) for the line termination.

This project mitigates multiple thermal constraints under NERC TPL-001-5 and increases capacity in the northeast area of Georgia in addition to improving overall voltage profile of the area under key 500kV contingencies and unit out generation scenarios. Solar sensitivity studies, **REDACTED** of solar penetration, showed that this new 500kV line increases capacity in transmission elements under solar expansion scenarios improving the transfer capability of generation in the area. This project will be done in conjunction with the GTC: East Walton-Middle Fork 500kV (TEAMS 21075).

Figure 12 Area System Map

REDACTED

Compliance Statement

This project addresses problems associated with Category P1 events. These problems were identified as part of Southern Company's Transmission Planning process in compliance with NERC Standard TPL-001-5.

Background and Problem Description

The McGrau Ford – Middle Fork 500kV in conjunction with the new GTC: East Walton-Middle Fork 500kV line project has multiple drivers: it alleviates thermal constraints, increases capacity in the area and provides voltage support for non-converge cases.

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Thermal Impacts:

The McGrau Ford-Middle Fork 500kV line effectively reduces flow and increases capacity on the circuits listed below. Table 25 denotes flow reductions in percent difference between flow currently on the circuits compared to when the McGrau Ford – Middle Fork 500kV project is included. Table 26 denotes increases in available capacity compared to the MVA rating of the circuit.

Table 25 Load Flow Reductions

Monitored Facility	Rating (MVA)	Base case, post-contingency loading (MVA)	Post-contingency loading with MGF-MF 500kV (MVA)	Flow reductions (% MVA)
South Hall 500/230kV LS Breaker #2	REDACTED	REDACTED	REDACTED	REDACTED
Norcross – South Hall 500kV	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500/230kV Bank A *	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #1	REDACTED	REDACTED	REDACTED	REDACTED
South Hall – Winder Primary 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Middle Fork – Unity 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Center Primary – Commerce Primary 115kV *	REDACTED	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Dawson Crossing – Nelson (White) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (White) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Dawson Crossing 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Bio – Pooles Creek 115kV *	REDACTED	REDACTED	REDACTED	REDACTED
Avalon Junction – Pooles Creek 115kV *	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (White) 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank C	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (Black) 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank D	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (Black) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank C	REDACTED	REDACTED	REDACTED	REDACTED
Athens – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Buford #3 - Suwanee	REDACTED	REDACTED	REDACTED	REDACTED
Shoal Creek 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Northwinds – Ocee 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Buford Dam – Shoal Creek 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – McEver Road 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Alpharetta – Northwinds 230kV	REDACTED	REDACTED	REDACTED	REDACTED
East Watkinsville 230/115kV Bank A *	REDACTED	REDACTED	REDACTED	REDACTED
Alpharetta – Glaze Drive 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Dawson Crossing – Nelson (Black) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Norcross – Ocee 230kV **	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank D	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank E	REDACTED	REDACTED	REDACTED	REDACTED

* Indicates constraint seen in 2024 V2B screen, addressed by McGrau Ford – Middle Fork 500kV project

**Note: Norcross-Ocee 230kV post MVA exceeds the line rating due to the voltage on the line being higher under contingency causing the MVA to jump above the rating. However, the post contingency amps are REDACTED whereas the circuit rating is REDACTED. The line loading is calculated off amps not the MVA, and the loading is at REDACTED.

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Table 26 Capacity Increases

Monitored Facility	Rating (MVA)	Increase in Available Capacity (% MVA)	Increase in Available Capacity (MVA)
Norcross – South Hall 500kV	REDACTED	REDACTED	REDACTED
South Hall 500/230kV Bank A	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #2	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #1	REDACTED	REDACTED	REDACTED
South Hall – Winder Primary 230kV	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank D	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank E	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 230kV	REDACTED	REDACTED	REDACTED
Dawson Crossing 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (White) 230kV	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank C	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (Black) 230kV	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank D	REDACTED	REDACTED	REDACTED
Bio – Pooles Creek 115kV	REDACTED	REDACTED	REDACTED
Avalon Junction – Pooles Creek 115kV	REDACTED	REDACTED	REDACTED
Alpharetta – Northwinds 230kV	REDACTED	REDACTED	REDACTED
Northwinds – Ocee 230kV	REDACTED	REDACTED	REDACTED
Center Primary – Commerce Primary 115kV	REDACTED	REDACTED	REDACTED
Alpharetta – Glaze Drive 230kV	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (White) 115kV	REDACTED	REDACTED	REDACTED
Norcross – Ocee 230kV	REDACTED	REDACTED	REDACTED
East Watkinsville 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Middle Fork – Unity 115kV	REDACTED	REDACTED	REDACTED
Dawson Crossing – Nelson (White) 115kV	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank C	REDACTED	REDACTED	REDACTED
Shoal Creek 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (Black) 115kV	REDACTED	REDACTED	REDACTED
Gainesville #2 – McEver Road 115kV	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED
Buford Dam – Shoal Creek 115kV	REDACTED	REDACTED	REDACTED
Athens – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED
Buford #3 – Suwanee	REDACTED	REDACTED	REDACTED
Dawson Crossing – Nelson (Black) 115kV	REDACTED	REDACTED	REDACTED

Voltage Impact:

In addition to alleviating thermal constraints and increasing capacity, the McGrau Ford-Middle Fork 500kV line provides VAR support to the northeast area. The McGrau Ford – Middle Fork 500kV project along with the GTC: East Walton – Middle Fork 500kV project alleviates the severe voltage depression issues seen in many different scenarios, due to the voltage support provided by the connection of the northern 500kV system. The base 2024 V2B cases had the stability-driven +150/-150 STATCOMs at McGrau Ford 230kV and Middle Fork 230kV placed online to alleviate non-converged case issues in later years.

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Table 27 indicates post-contingency voltages in worst case scenarios, and equivalent amount of reactive support required at South Hall 500kV indicated as comparison to the level of voltage support that the McGrau Ford – Middle Fork 500kV project provides to the area.

Table 27 Post-contingency Voltages in Worst Case Scenarios

Case	Post-Contingency Voltage (PU)	Voltage with McGrau Ford – Middle Fork 500kV (PU)	Equivalent amount of reactive support (MVAR)
REDACTED	REDACTED	REDACTED	REDACTED
REDACTED	REDACTED	REDACTED	REDACTED
REDACTED	REDACTED	REDACTED	REDACTED
REDACTED	REDACTED	REDACTED	REDACTED

The results in tables 28 and 29 show the post-contingency voltage results for non-converge base cases versus the McGrau Ford-Middle Fork 500kV line being in service with the mentioned STATCOMs OFF and ON, respectively.

Table 28 provides the voltage comparison of the North GA 500kV system in the version 2B load flow cases with the McGrau Ford – Middle Fork 500kV line in service and stability STATCOMs off:

Table 28 Voltage Results STATCOM OFF

Bus	Case	Base V2B Post-Contingency Voltage (PU) / (kV)		Post Contingency Voltage with McGrau Ford – Middle Fork 500kV (PU) / (kV)		Difference (PU) / (kV)	
Big Shanty 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Bull Sluice 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Klondike 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Ohara 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Ashley Park 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Dresden 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Union City 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Villa Rica 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
McGrau Ford 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
East Walton 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED

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Figures 13 through 16 show the post-contingency voltage for base cases vs the McGrau Ford-Middle Fork 500kV line being in service and the STATCOMs off for several 500kV facilities in North Georgia.

Figure 13 Voltage Profile-Norcross-STATCOM Off

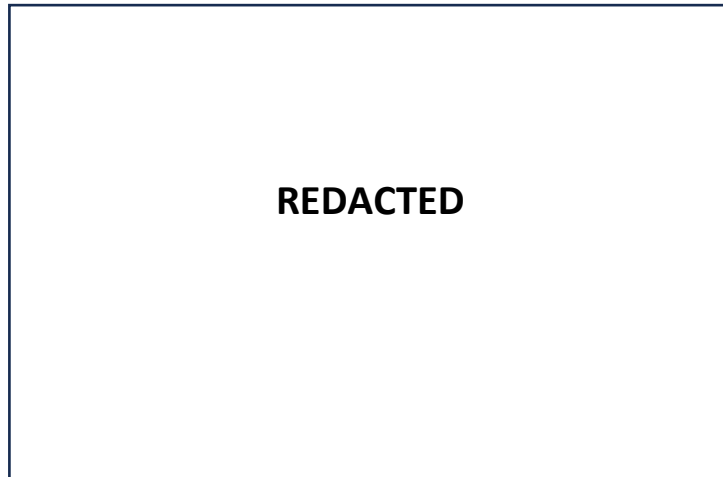


Figure 14 Voltage Profile-Villa Rica-STATCOM Off

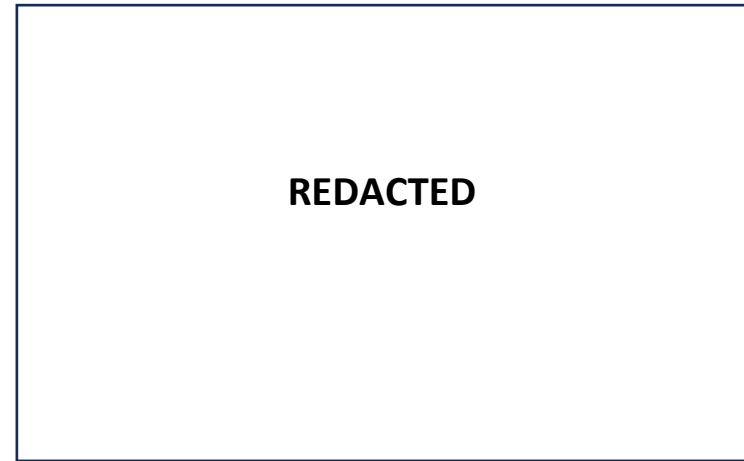


Figure 15 Voltage Profile-South Hall-STATCOM Off

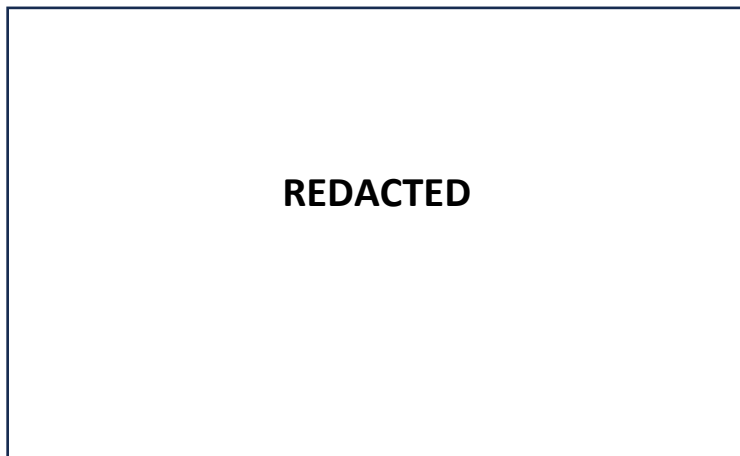
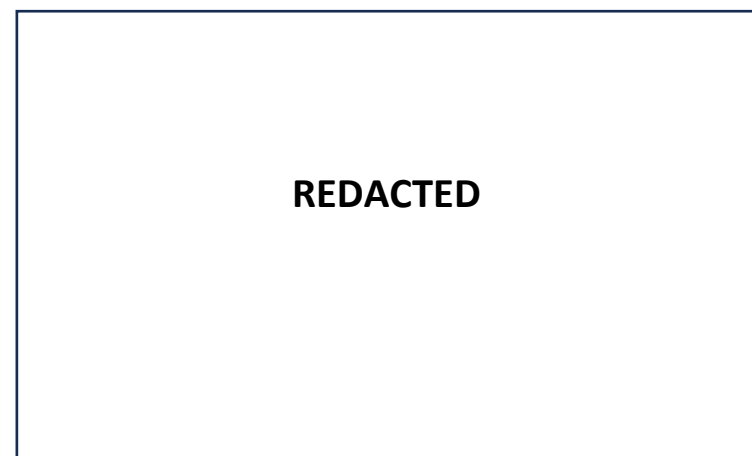


Figure 16 Voltage Profile-McGrau Ford-STATCOM Off



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Table 29 provides the voltage comparison of the North GA 500kV system in the version 2B load flow base cases with the McGrau Ford – Middle Fork 500kV line in service with stability STATCOMs on:

Table 29 Voltage Results STATCOM ON

Bus	Case	Base V2B Post-Contingency Voltage (PU) / (kV)		Post Contingency Voltage with McGrau Ford – Middle Fork 500kV (PU) / (kV)		Difference (PU) / (kV)	
Big Shanty 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Bull Sluice 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Klondike 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Ohara 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Ashley Park 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Dresden 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Union City 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
Villa Rica 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
McGrau Ford 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500kV	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
East Walton 500kV	REDACTED REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED
	REDACTED REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED	REDACTED

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Figures 17 through 20 show the post-contingency voltage for base cases vs the McGrau Ford-Middle Fork 500kV line being in service and the STATCOMs on for several 500kV facilities in North Georgia.

Figure 17 Voltage Profile-Villa Rica-STATCOM ON

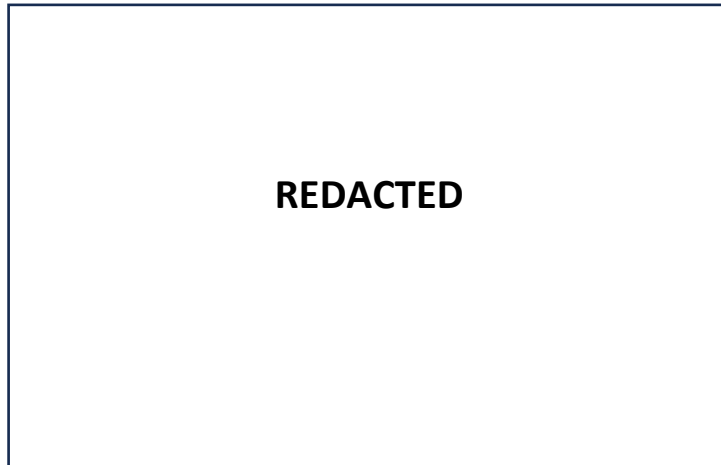


Figure 18 Voltage Profile-Norcross-STATCOM Off

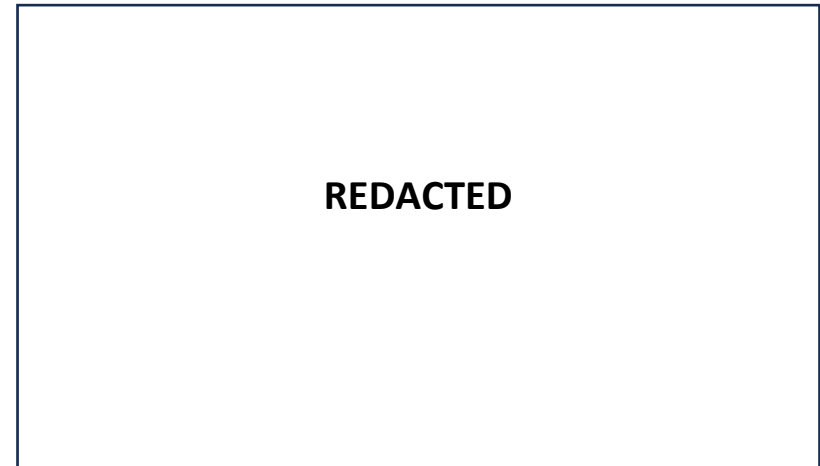


Figure 19 Voltage Profile-McGrau Ford-STATCOM Off

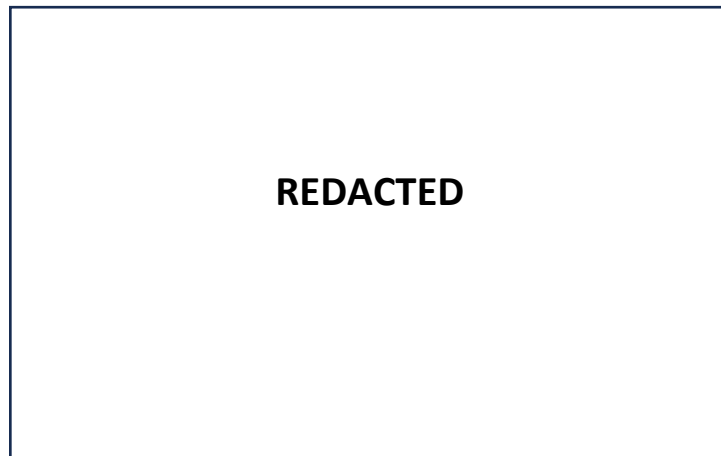
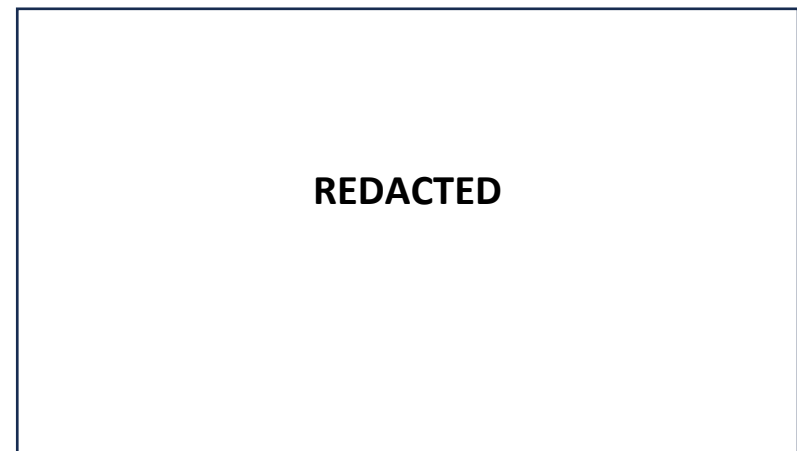


Figure 20 Voltage Profile-South Hall-STATCOM Off



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Figures 21 through 24 show the improved voltage profile on key buses with all new 500kV projects in place.

Figure 21 Voltage Profile-Norcross

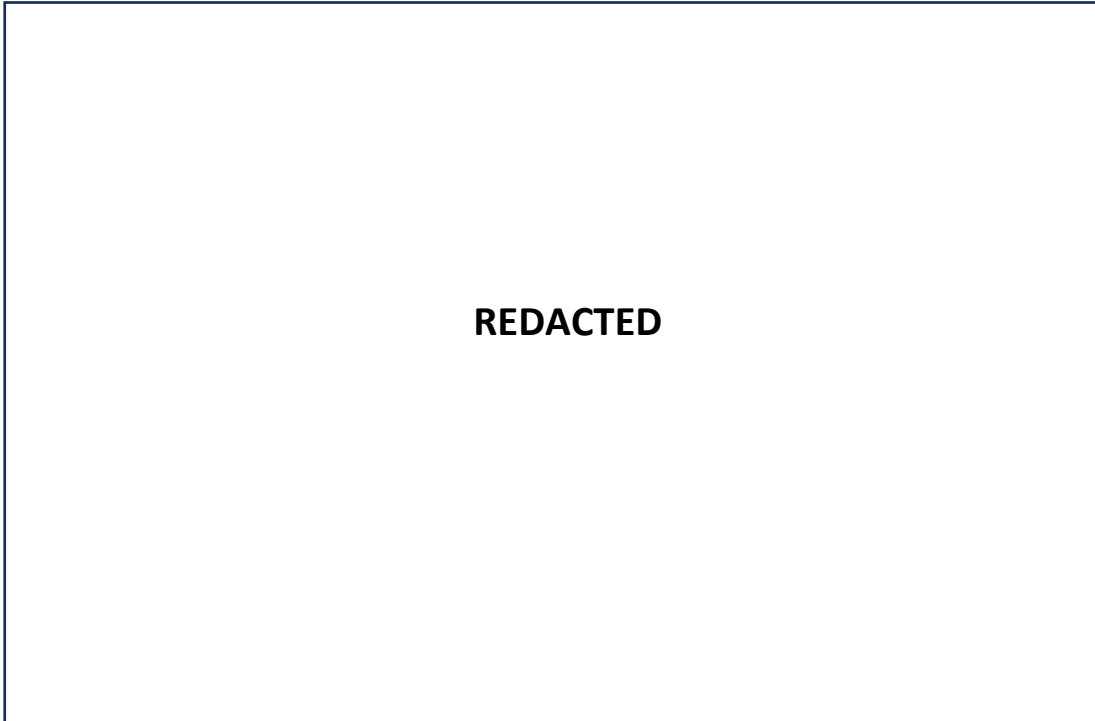
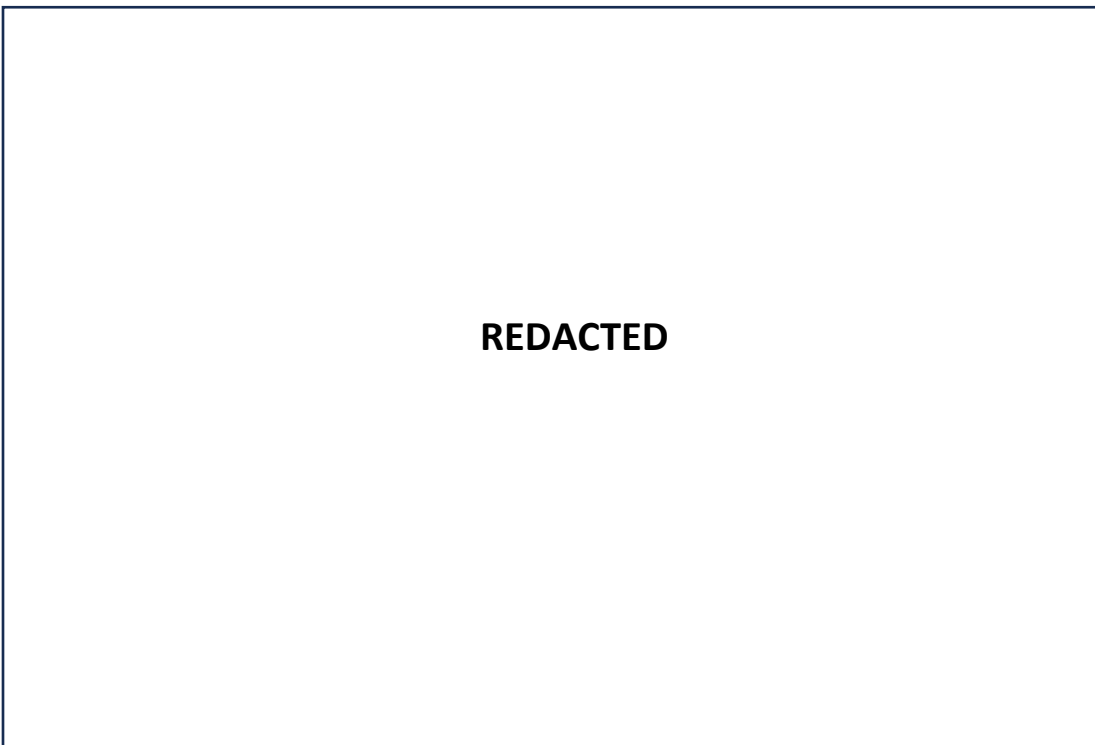


Figure 22 Voltage Profile-McGrau Ford



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Figure 23 Voltage Profile-East Walton

REDACTED

Figure 24 Voltage Profile-Rockville

REDACTED

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Solar Sensitivity Analysis

Studies were performed to analyze the increase of capacity and flow reduction in the area with the McGrau Ford-Middle Fork 500kV line in service for Solar Sensitivity cases. Table 30 shows the loading of several lines and elements in the northeast area for solar sensitivity cases with the line out of service and in service. Table 31 shows the increase of capacity for these lines and elements with the line in service.

Table 30 Load Flow Reductions – Solar Sensitivity

Monitored Facility	Rating (MVA)	SS Base case, post-contingency loading (MVA)	SS post-contingency loading with MGF-MF 500kV (MVA)	Flow reductions (% MVA)
South Hall 500/230kV LS Breaker #2	REDACTED	REDACTED	REDACTED	REDACTED
Norcross – South Hall 500kV	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500/230kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #1	REDACTED	REDACTED	REDACTED	REDACTED
South Hall – Winder Primary 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Bull Sluice - Norcross 500kV	REDACTED	REDACTED	REDACTED	REDACTED
Middle Fork – Unity 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Center Primary – Commerce Primary 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (White) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Glaze Drive – Norcross 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Carters Dam – Nelson 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank C	REDACTED	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (White) 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (Black) 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank C	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank D	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (Black) 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Buford Dam – Shoal Creek 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Dawson Crossing 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Buford #3 - Suwanee	REDACTED	REDACTED	REDACTED	REDACTED
Shoal Creek 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Gainesville #2 – McEver Road 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Northwinds – Ocee 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Alpharetta – Northwinds 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED	REDACTED
Norcross – Ocee 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Shoal Creek – South Hall 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Alpharetta – Glaze Drive 230kV	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank D	REDACTED	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank E	REDACTED	REDACTED	REDACTED	REDACTED
East Watkinsville 230/115kV Bank A	REDACTED	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 230kV	REDACTED	REDACTED	REDACTED	REDACTED

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Table 31 Capacity Increase – Solar Sensitivity

Monitored Facility	Rating (MVA)	Increase in Available Capacity (% MVA)	Increase in Available Capacity (MVA)
Norcross – South Hall 500kV	REDACTED	REDACTED	REDACTED
South Hall 500/230kV Bank A	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #2	REDACTED	REDACTED	REDACTED
South Hall 500/230kV LS Breaker #1	REDACTED	REDACTED	REDACTED
Bull Sluice - Norcross 500kV	REDACTED	REDACTED	REDACTED
South Hall – Winder Primary 230kV	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank D	REDACTED	REDACTED	REDACTED
Norcross 500/230kV Bank E	REDACTED	REDACTED	REDACTED
Glaze Drive – Norcross 230kV	REDACTED	REDACTED	REDACTED
Carters Dam – Nelson 230kV	REDACTED	REDACTED	REDACTED
Shoal Creek – South Hall 230kV	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (White) 230kV	REDACTED	REDACTED	REDACTED
Gainesville #2 – South Hall (Black) 230kV	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank C	REDACTED	REDACTED	REDACTED
Gainesville #2 230/115kV Bank D	REDACTED	REDACTED	REDACTED
Norcross – Ocee 230kV	REDACTED	REDACTED	REDACTED
Alpharetta – Northwinds 230kV	REDACTED	REDACTED	REDACTED
Northwinds – Ocee 230kV	REDACTED	REDACTED	REDACTED
Alpharetta – Glaze Drive 230kV	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (White) 115kV	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank C	REDACTED	REDACTED	REDACTED
Bio 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Shoal Creek 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Center Primary – Commerce Primary 115kV	REDACTED	REDACTED	REDACTED
Dawson Crossing 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Gainesville #1 – Gainesville #2 (Black) 115kV	REDACTED	REDACTED	REDACTED
Gainesville #2 – McEver Road 115kV	REDACTED	REDACTED	REDACTED
Middle Fork – Unity 115kV	REDACTED	REDACTED	REDACTED
East Watkinsville 230/115kV Bank A	REDACTED	REDACTED	REDACTED
Buford Dam – Shoal Creek 115kV	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 230kV	REDACTED	REDACTED	REDACTED
Buford #3 - Suwanee	REDACTED	REDACTED	REDACTED
Athena – East Watkinsville 115kV	REDACTED	REDACTED	REDACTED

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Study Assumptions

Base Case Thermal Study

- Load Flow Cases: 2024 series v2b
- Study Years: 2031-2034
- Case Seasons: Summer Peak, Off-Peak, Daylight Shoulder, Dusk Shoulder

Voltage Comparison Study

- Load Flow Cases: 2024 series v2b
- Study Years: 2033-2034
- Case Seasons: Summer Peak - Only non-converged base cases were evaluated

Solar Sensitivity Thermal Study

- Load Flow Cases: 2024 series v2b
- Study Years: 2031-2034
- Solar Penetration: **REDACTED**
- Case Seasons: Summer Peak, Off-Peak, Daylight Shoulder, Dusk Shoulder

Discussion of Alternatives

The Preferred Plan (\$REDACTED) includes the following:

- Construction of the McGrau Ford-Middle Fork 500 kV Line (approximately 65 miles) with (3) 100C 1113 ACSR conductor.
- Expand the McGrau Ford 500kV yard and convert to a ring bus and build a 500kV yard at Middle Fork to terminate the new line.
- Installation of communication equipment and integration with EMS.

The preferred plan accomplishes the following:

1. **High-Capacity Corridor:** Establishes a new 500 kV line from McGrau Ford to Middle Fork, connecting the 500kV system in the north of Georgia enhancing the system capacity.
2. **Reduced Flows and Increased Capacity:** Decreases flows and increases available capacity on multiple circuits, improving system reliability in the northeast.
3. **Supports Generation Expansion:** Facilitates integration of additional generation at existing facilities along with new solar facilities, promotes sustainability, and grid resilience.
4. **Operational Flexibility:** Provides significantly improved operational flexibility and reliability for system outages and future maintenance activities on existing facilities, particularly on the critical 500 kV system.

Alternate Plans

- **Alternate Plan 1:** Do nothing (\$REDACTED). This is not a viable option to remain in compliance with NERC TPL-001-5. In addition, the trend of increasing penetration of renewable (solar) generation plants on the Georgia Integrated Transmission System (ITS), and the resulting displacement of traditional fossil-fueled resources (particularly older coal facilities) is changing flow patterns on the system. These changes in addition to evolving interregional flow patterns and forecasted load growth are resulting in transmission congestion. If left unaddressed the system will not be able to facilitate future growth and maintain reliability requirements. This will introduce potential delays and limits until transmission improvements are identified and completed on an individual basis at the time of interconnection.

PUBLIC DISCLOSURE

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- **Alternate Plan 2:** Brute Force (Total of \$REDACTED). This involves the direct replacement or rebuild of the facilities noted in Table 32.

Table 32 Brute Force Estimates

Facility Name	Rebuild Line Length (Miles)	Planning Grade Estimate
Avalon Junction – Pooles Creek – Bio 115kV rebuild	20.5	REDACTED
Center Primary – Commerce Primary 115kV rebuild	17	REDACTED
East Watkinsville 230/115kV new auto transformer	N/A	REDACTED
Norcross – Ocee 230kV rebuild	3.5	REDACTED
South Hall 500/230kV new auto transformer	N/A	REDACTED
STATCOMs Installation for VAR Support	N/A	REDACTED
Total	41	REDACTED

Alternate Plan 2 is not a feasible option given the following factors:

1. Accommodating and coordinating necessary outages to facilitate maintenance of existing facilities and the construction activities for system improvements
2. Timing and reduced reliability exposure to system given length of corridors
3. Lack of solution robustness to facilitate power transfers from south Georgia to north Georgia driven by increased solar generation and load growth

Additionally, the Brute Force alternative estimates do not reflect the full scope of all necessary work and costs that would be required along with any additional generation redispatch impacts associated with the numerous required outages.

Conclusion and Recommendations

Although the Preferred Plan is not the least cost plan, it was selected because it solves identified thermal constraints, provides additional 500kV corridor for generation power transfers, and provides voltage support to the area and for non-converging cases.